

1. O/N/23/21/Q9

The planet Venus orbits the Sun at a constant speed of $3.5 \times 10^4 \text{ m/s}$ and takes a time T_V to complete one orbit.

- (a) Venus is always $1.1 \times 10^{11} \text{ m}$ from the Sun.

Calculate T_V .

$T_V = \dots\dots\dots \text{ s}$ [2]

- (b) Explain why speed is a scalar quantity.

.....
..... [1]

- (c) Velocity is a vector quantity.

- (i) Explain what happens to the velocity of Venus as it orbits the Sun.

.....
.....
.....
..... [2]

- (ii) There is a resultant force on Venus as it orbits the Sun. The force is perpendicular to the direction of the motion of Venus.

State why this force is needed.

.....
..... [1]

(iii) Fig. 9.1 shows the orbit of Venus around the Sun.

On Fig. 9.1, draw an arrow to show the resultant force on Venus.

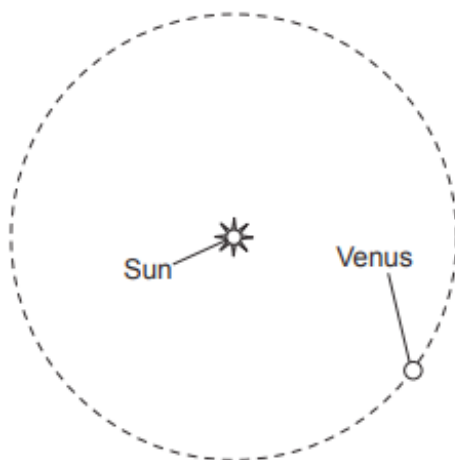


Fig. 9.1

[1]

(iv) State what causes the force acting on Venus.

.....
.....
..... [2]

(d) One planet in the Solar System is closer to the Sun than Venus.

(i) State the name of this planet.

..... [1]

(ii) Compare the time that this planet takes to complete one orbit of the Sun with T_V and explain the difference.

.....
.....
.....
..... [2]

[Total: 12]

2. M/J/23/22/Q10

(a) Astronomical distances are measured in light-years.

(i) State what is meant by 'a light-year'.

.....
 [1]

(ii) The Sun is one star in the Milky Way galaxy.

State the approximate diameter of the Milky Way galaxy.

diameter of Milky Way = light-years [1]

(b) There are several stages in the life cycle of a star.

(i) Complete Fig. 10.1 to show the stages that a **massive** star goes through after it has used up most of the hydrogen at the centre of the star.

Use words from the following list:

nebula neutron star protostar red giant supernova white dwarf

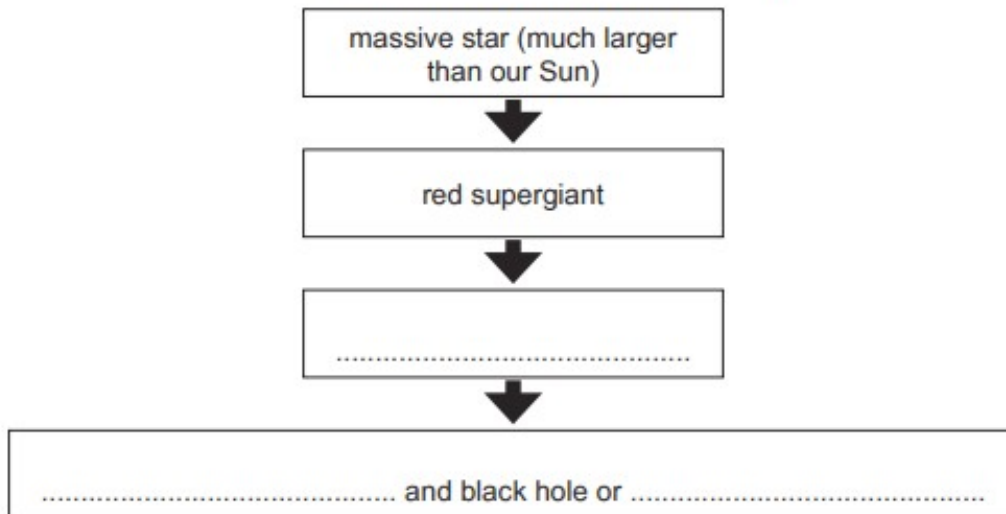


Fig. 10.1

[2]

(ii) State the stage in the life cycle of a star where heavy elements are formed.

..... [1]

- (c) Current scientific understanding is that the universe began 14 billion years ago in an event known as the Big Bang.

Explain **one** observation that supports the Big Bang Theory.

observation

.....

explanation

.....

.....

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.....

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[4]

[Total: 9]